

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**B. Tech. EE Sem-IV Theory Examination****(April-May, 2018)****EE 206 Digital Electronics and Microprocessor****Date: 04/05/2018, Friday****Time: 10:00 a.m. to 01:00 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Comments are necessary in each program.

SECTION-1**Q-1 Answer the question below.****[7]**

1. Find binary number for $(56)_{10}$.
2. $A+A' =$ _____.
3. $(A3)_{16} = (\text{____})_{10}$
4. A demultiplexer is also known as _____.
5. The digital code ASCII stand for _____.
6. In logic circuit diagrams, a small circle or bubble represents _____.
(a) Addition(OR) (b) Multiplication (AND)
(c) Inversion (NOT) (d) None of these.
7. The base used for octal number system is _____.

Q-2(a) What is half sub tractor? Draw the logic diagram and truth table of half sub tractor.**[4]****Q-2(b) Answer any two questions.****[10]**

1. Draw the circuit diagram of a positive level triggered J-K Flip Flop and explain its operation with the help of a truth table.
2. What is meant by multiplexer? Write the advantages of multiplexers.
3. Perform $(9)_{10} - (4)_{10}$ using 1's complement method.

Q-3(a) Answer any two questions.**[10]**

1. State and explain 2-input CMOS NAND gate in detail.
2. Write notes on BCD to 7 segment decoders.
3. Explain the working of a full adder using two half adder circuit with its truth table

Q-3(b) If $A=1$, $B=0$ and $C=1$, find the values of**[4]**

1. $(A'B') + (AB) + B' + C$
2. $AB + BC + AC + ABC$

SECTION-2**Q-4 Answer the question below.****[7]**

1. A microprocessor is a ____ bit device.
2. A Stack Pointer is a ____ bit register.
3. Convert 1110 0001 to Decimal number.
4. Byte is a group of ____ bits.
5. _____ is non-maskable interrupt in 8085.

6. An accumulator is ____ bit register.

7. How many address and data lines are available in 8085 microprocessor?

Q-5(a) Draw pin diagram of 8085 and explain function of INTA and INTR. [5]

OR

Q-5(a) Explain 8085 De-multiplexed Address and Data Bus with control signals with necessary diagram. [5]

Q-5(b) Draw structure of Flag register and explain each bit of it. [5]

Q-5(c) Draw timing diagram for execution sequence of instruction MVI A,25H written at memory location 2000H. [4]

OR

Q-5(c) Explain various addressing modes of 8085. [4]

Q-6(a) Draw functional block diagram of 8085 and explain Stack Pointer(SP) and Program Counter in detail(PC). [8]

OR

Q-6(a) Draw memory-interfacing diagram to interface 2kB RAM and 2kB of ROM with 8085. Starting address of RAM is 0000H and ROM address start at the end of RAM. [8]

Q-6(b) Write an assembly level program (attempt any two) [6]

1. To add two 8-bit numbers stored at memory location 5000H and 5001H, store the answer in next memory location
2. Write an assembly level program to find out 2's complement of number stored at memory location 3000H and store answer at memory location 3001H.
3. To generate 100micro second delay, consider crystal frequency of 3MHz.
4. To multiply two numbers stored in register B and C, store answer in register D.
